How machines learn from examples



AI · Level 1 · Lesson 2

Today we will

- Put the five steps of machine learning in order.
- Find a hidden pattern in labelled examples and use it to predict new ones.
- Explain how a spam filter learns from labelled messages.

Remember

- People collect examples and label them. Training finds patterns in this training data and stores them as a model.
- We test the model on examples it has never seen. More, and more varied, examples usually help.
- The model makes predictions — about now (“Is this spam?”) or the future. Many models stay the same until people retrain them.

Be the machine: Zog game

1. Warm-up (5 min): Draw a ripe and an unripe mango on the board (or hold up real ones).
2. Teach (8 min): Draw the five steps with simple icons: collect examples → label them → train: find patterns → test on new examples → use: make predictions.
3. Zog game (14 min): Draw 6 zogs and 6 not zogs in two columns.
4. Pair talk (7 min): Pairs share the pattern they found.
5. Wrap-up (6 min): Link the game to the five steps: the labelled drawings were the training data, and the new shapes were the test.

Secret pattern: zogs have three corners. Plan 6 zogs, 6 not zogs (squares, circles, stars) and 6 test shapes, including an upside-down triangle. Vary size, shading and direction in both groups, so only the corners matter.

Let’s discuss

1. To teach a young child which mangoes are ripe, would you give a rule, or show many mangoes and say ‘ripe’ or ‘not ripe’?
2. Which examples helped you most?
3. What new examples would help a machine that got this wrong?

Exit ticket

- Write the five steps in order, and one thing a spam filter learns from.